with a bang. But there are a number of other features inherent to all inflationary models that are vital because they solve important problems that have stumped standard big bang cosmology.

Inflation and the Horizon Problem

One such problem is called the *horizon problem* and concerns the uniformity of the microwave background radiation that we came across previously. Recall that the temperature of the microwave radiation reaching us from one direction in space agrees with that coming from any other direction to fantastic accuracy (to better than a thousandth of a degree). This observational fact is pivotal, because it attests to homogeneity throughout space, allowing for enormous simplifications in theoretical models of the cosmos. In earlier chapters, we used this homogeneity to narrow down drastically the possible shapes for space and to argue for a uniform cosmic time. The problem arises when we try to explain *how* the universe became so uniform. How is it that vastly distant regions of the universe have arranged themselves to have nearly identical temperatures?

If you think back to Chapter 4, one possibility is that just as nonlocal quantum entanglement can correlate the spins of two widely separated particles, maybe it can also correlate the temperatures of two widely separated regions of space. While this is an interesting suggestion, the tremendous dilution of entanglement in all but the most controlled settings, as discussed at the end of that chapter, essentially rules it out. Okay, perhaps there is a simpler explanation. Maybe a long time ago when every region of space was nearer to every other, their temperatures equalized through their close contact much as a hot kitchen and a cool living room come to the same temperature when a door between them is opened for a while. In the standard big bang theory, though, this explanation also fails. Here’s one way to think about it.

Imagine watching a film that depicts the full course of cosmic evolution from the beginning until today. Pause the film at some arbitrary moment and ask yourself: Could two particular regions of space, like the kitchen and the living room, have influenced each other’s temperature? Could they have exchanged light and heat? The answer